# Week 31: Hidden orchard

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Each problem's key repeats the main answer from the print guide, week-31-facilitator.pdf, whose source is editable/facilitator-src/guide.json; the hints and the converse argument are on the guide page named in the key's first line. The overview repeats the guide's first page. Change both together; check_guide_copy.py checks that every sentence here is in the guide.

# Mathematical overview

## The mathematics in one view

A nonzero lattice point $T = (a, b)$ is **visible** from $O = (0, 0)$ when the open line segment $OT$ contains no other lattice point. A blocker must lie exactly on that segment; O and T are not blockers. The packet uses the first quadrant, including the axes.

**Exact criterion.** For nonnegative integer coordinates, not both zero, let $d = \gcd(a, b)$. The first lattice point on the ray is $(a/d, b/d)$, and the interior points are exactly $k(a/d, b/d)$, for $k = 1, \ldots, d - 1$. Thus T is visible if and only if $d = 1$, and there are $d - 1$ blockers. On an axis use $\gcd(a, 0) = a$; only the first unit point is visible.

**What the geometry encodes.** Every lattice ray has one primitive first point, with coprime coordinates. Its other lattice points are positive integer multiples of that point. Scaling a direction explains hidden dots; a common divisor measures how many equal primitive steps reach the target.

**Limits.** These are ideal points, not thick trees: an almost-collinear dot does not block. On these rectangular grids every intermediate lattice point is present. Visibility could differ if dots were removed. The source's asymptotic density discussion is not part of this lesson.

## What each group is doing

**K–1:** Test exact sight lines, find first blockers, and compare rows and rays. Counting to 6 and holding a thread straight suffice; coordinates may be spoken or pointed to. A demonstration of a blocker is a valid explanation of a particular hidden target.

**Grades 2–3:** Group dots sharing a direction and conjecture a divisibility rule from examples, then explain a fully visible row. Coordinates mean "across, up"; small equal groups and multiplication are helpful.

**Grades 4–5:** Explain both directions of the visibility criterion and why the blocker list is complete. Division, common factors and proportional steps are the prerequisites. The reduced-fraction argument can stay with the facilitator if children can only justify particular rays.

# Problems and routes

| Problem | K–1 | 2–3 | 4–5 | route |
|---|---|---|---|---|
| Every dot visible from O | P1 | P1 | P1 | first |
| Test given targets for blockers | P2 | P2 | P2 | first |
| Every target with exactly one or exactly two blockers | P3 | P3 | | return |
| Predict the first dot and the blockers of far targets | | | P3 | return |
| A row of visible dots, and a row of hidden dots? | P4 | | | return |
| Groups of dots on one straight line from O | | P4 | P4 | return |
| Common factors and blockers, both directions | | | P5 | return |
| Which visible dot hides the most dots | P5 | | | return |
| A number rule for visible dots | | P5 | | return |
| A row where every dot is visible | | P6 | | return |
| Why the rule finds every blocker | | | P6 | return |
| A visible row, an infinite hidden set, 99 blockers | | | P7 | return |

**First route.** K–1 P1–P2; Grades 2–3 P1–P2; Grades 4–5 P1–P2. **Return visits.** K–1 P3–P5; Grades 2–3 P3–P6; Grades 4–5 P3–P7.

# Key, one entry per problem

## Every dot visible from O

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

*Idea.* A dot (a, b) is visible from O exactly when a and b have no common factor larger than 1: 13 dots on the 0-4 grid, 25 on the 0-6 grid.

**K–1:** *Print guide, p. 2.* Each row gives the across-coordinates x of *all* visible dots at height y. Exclude O. The K grid has 13 visible dots; the 0-6 grid has 25.

| Height y | K grid: x=0...4 | Larger grid: x=0...6 |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 0,1,2,3,4 | 0,1,2,3,4,5,6 |
| 2 | 1,3 | 1,3,5 |
| 3 | 1,2,4 | 1,2,4,5 |
| 4 | 1,3 | 1,3,5 |
| 5 | Outside this grid | 1,2,3,4,6 |
| 6 | Outside this grid | 1,5 |

**Grades 2–3:** *Print guide, p. 3.* Use the complete catalog on page 2. Three possible first-blocker marks: (6,4) → (3,2), (6,3) → (2,1), (4,4) → (1,1). Accept any three correct hidden targets.

**Grades 4–5:** *Print guide, p. 4.* The 25 visible dots are in the catalog on page 2. A hidden target has one blocker fewer than the gcd of its coordinates, and on this grid the gcd is at most 6, so the possible counts are 1, 2, 3, 4 and 5. The diagonal shows them all: (2,2) has 1 blocker, (3,3) has 2, (4,4) has 3, (5,5) has 4 and (6,6) has 5. The axes work the same way, and (6,4) or (6,3) give 1 or 2 off the diagonal. Other correct choices are welcome.

## Test given targets for blockers

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

*Idea.* The blockers of (a, b) are the multiples k(a/d, b/d) for k = 1, ..., d - 1, where d is the greatest common factor; the first is the nearest to O.

**K–1:** *Print guide, p. 2.* First picture: targets (4,4), (4,2), (2,4), (3,3) are all hidden; their nearest blockers are (1,1),

(2,1), (1,2), (1,1), respectively. Second picture: (4,0) is blocked first by (1,0); (0,4) by (0,1); (2,2) by (1,1); (4,3) is visible and has no blocker.

**Grades 2–3:** *Print guide, p. 3.* The requested complete interior lists are below. The target itself never goes in the blocker list.

| Target | All blockers between O and target |
|---|---|
| (6,4) | (3,2) |
| (6,3) | (2,1), (4,2) |
| (5,3) | None |
| (5,2) | None |
| (4,4) | (1,1), (2,2), (3,3) |
| (4,1) | None |

**Grades 4–5:** *Print guide, p. 4.* For (6,4), first (3,2), only blocker (3,2). For (6,3), first (2,1), blockers (2,1), (4,2). For (5,3), first (5,3) itself, no blockers. For (6,6), first (1,1), blockers (k,k), k=1,...,5. For (6,0), first (1,0), blockers (k,0), k=1,...,5. For (0,6), first (0,1), blockers (0,k), k=1,...,5.

## Every target with exactly one or exactly two blockers

K–1 P3 · Grades 2–3 P3 · return route · wording differs by band

*Idea.* On the 0-6 grid, exactly one blocker means a greatest common factor of 2 (nine targets), exactly two means 3 (five targets).

**K–1:** *Print guide, p. 2.* Exactly one blocker: (0,2), (2,0), (2,2), (2,4), (2,6), (4,2), (4,6), (6,2), (6,4). Exactly two: (0,3), (3,0), (3,3), (3,6), (6,3). Their gcds are 2 and 3 respectively. Use thread to check; children need not write these coordinate lists. An adult can write on the lines, or a child who does not write can mark the two kinds of target in two colours on the grid.

**Grades 2–3:** *Print guide, p. 3.* The two complete target lists are exactly K P3 on page 2, including axis points. “One blocker” means gcd 2, not just “both coordinates even”: (4,4) has three blockers and (6,6) has five.

## Predict the first dot and the blockers of far targets

Grades 4–5 P3 · return route

*Idea.* The first dot of (a, b) is (a/d, b/d) and there are d - 1 blockers, d the greatest common factor; (12, 8), (15, 10), (21, 14) all lie on the ray of (3, 2).

*Print guide, p. 4.* Targets (12,8), (15,10), (21,14) all have first point (3,2) and respectively 3,4,6 blockers. For (25,15), first (5,3), four blockers. For (13,8), first (13,8), zero blockers. The rule applies to every nonzero whole-coordinate target; O itself has no direction and is excluded. The extra grid on that page runs to (25,15) at 7 mm, so each of these targets can be checked with thread; about 20 cm reaches the farthest.

## A row of visible dots, and a row of hidden dots?

K–1 P4 · return route

*Idea.* Row 1 has only visible dots; no row has only hidden dots, since (1, y) is visible in every row.

*Print guide, p. 2.* Row y=1 is the only row where every dot is visible, from (0,1) to (6,1). Row 0 has the hidden dot (2,0), and each row y=2,...,6 has the hidden dot (0,y). No row is all hidden, since (1,y) is visible in every row, including (1,0). A child who turns the board may find column x=1, which is also all visible.

## Groups of dots on one straight line from O

Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

*Idea.* The dots on one ray from O are the multiples of its first dot, so every target belongs to exactly one group.

**Grades 2–3:** *Print guide, p. 3.* Excluding O, the five groups on the 0-6 grid are:
(1,1): (1,1), (2,2), (3,3), (4,4), (5,5), (6,6).
(2,1): (2,1), (4,2), (6,3).
(3,2): (3,2), (6,4).
(1,2): (1,2), (2,4), (3,6).
(1,0): (1,0), (2,0), (3,0), (4,0), (5,0), (6,0).
These are all positive multiples of the named first point that fit in the grid.

**Grades 4–5:** *Print guide, p. 4.* For (6,4), the group is {(3,2), (6,4)}. For (6,3), it is {(2,1), (4,2), (6,3)}. For (6,6), it is {(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)}. Each nonzero target determines one ray and that ray has one first lattice point, so the groups are disjoint.

## Common factors and blockers, both directions

Grades 4–5 P5 · return route

*Idea.* A common factor q > 1 gives the blocker (a/q, b/q); conversely a blocker t(a, b) with t = r/s in lowest terms forces s > 1 to divide both coordinates.

*Print guide, p. 4.* If $q > 1$ divides a and b, then (a/q,b/q) is a lattice point strictly between O and (a,b). Conversely, a blocker (u,v)=t(a,b) has $0 < t < 1$. Write t=r/s in lowest terms. Since ra/s and rb/s are integers and gcd(r,s)=1, s divides both a and b; also $s > 1$. Thus every blocker certifies a nontrivial common divisor. Do not assume every blocker is the first one or that its coordinates divide the target coordinates.

## Which visible dot hides the most dots

K–1 P5 · return route

*Idea.* On the 0-6 grid, (1, 0), (0, 1) and (1, 1) each hide five dots; every other visible dot hides at most two.

*Print guide, p. 2.* Three visible dots tie: (1,0), (0,1), (1,1). Each hides five dots, its multiples 2 through 6. Every other visible dot has some coordinate at least 2, so it reaches at most three dots within this grid and hides at most two.

## A number rule for visible dots

Grades 2–3 P5 · return route

*Idea.* A dot is visible exactly when its coordinates have no common factor larger than 1; dividing by a common factor gives a smaller step along the same thread.

*Print guide, p. 3.* A dot is visible exactly when the two coordinates have no common divisor greater than 1. Thus (12,8) is hidden (first point (3,2), three blockers); (10,7) is visible; (15,5) is hidden (first (3,1), four blockers); (11,1) is visible. Dividing both coordinates by a common divisor gives a smaller step in the same straight direction. The converse needs the reasoning on page 4.

## A row where every dot is visible

Grades 2–3 P6 · return route

*Idea.* Row 1 is the only row where every dot is visible: a point between O and (x, 1) would have height strictly between 0 and 1.

*Print guide, p. 3.* Row y=1: every dot (x,1), $x \geq 0$, is visible. Every interior point of the segment to (x,1) has height strictly between 0 and 1, so cannot be a lattice point. This is an exact geometric proof, not an inference from a finite row. It is the only such row: row 0 has the hidden dot (2,0), and each row $y \geq 2$ has the hidden dot (0,y). Accept column x=1 as well.

## Why the rule finds every blocker

Grades 4–5 P6 · return route

*Idea.* With u = a/d and v = b/d coprime, every lattice point on the ray is an integer multiple of (u, v), so the d - 1 multiples are all the blockers.

*Print guide, p. 4.* Put u=a/d, v=b/d; then gcd(u,v)=1. The d-1 points k(u,v), $1 \leq k < d$, are blockers. Any lattice point on that ray is t(u,v). For a positive coordinate, t is rational; write it in lowest terms. Its denominator must divide both u and v, hence is 1. Therefore t is an integer, so no extra points fit between consecutive listed points. This also covers axes: the primitive coordinate pair is (1,0) or (0,1).

## A visible row, an infinite hidden set, 99 blockers

Grades 4–5 P7 · return route

*Idea.* Row 1 is entirely visible, the diagonal (n, n) for n at least 2 is hidden behind (1, 1), and (100, 100) has exactly 99 blockers.

*Print guide, p. 4.* Row y=1 is fully visible by the height argument on page 3. The points (n,n), $n \geq 2$, form an infinite hidden collection, all behind (1,1). Target (100,100) has exactly 99 blockers, (k,k) for k=1,...,99; (100,0) is another valid example.